

## **PING(ICMP)- Task 4**

### **Aim**

To understand how ICMP (Internet Control Message Protocol) works using the ping command and analyze network connectivity between systems.

### **What is ICMP?**

ICMP is a network layer protocol used for:

- Error reporting
- Network diagnostics
- Connectivity testing

The ping command uses ICMP Echo Request and Echo Reply packets.

### **What is Ping?**

ping is a command-line utility used to check whether a host/device is reachable over a network.

It measures:

- Connectivity status
- Round Trip Time (RTT)
- Packet loss

### **Commands Used**

```
ping google.com
```

```
ping 192.168.1.254
```

```
cmnatic@CMNatic-THM-LPTOP:~$ ping 192.168.1.254
PING 192.168.1.254 (192.168.1.254) 56(84) bytes of data.
64 bytes from 192.168.1.254: icmp_seq=1 ttl=63 time=2.18 ms
64 bytes from 192.168.1.254: icmp_seq=2 ttl=63 time=2.53 ms
64 bytes from 192.168.1.254: icmp_seq=3 ttl=63 time=2.15 ms
64 bytes from 192.168.1.254: icmp_seq=4 ttl=63 time=3.34 ms
64 bytes from 192.168.1.254: icmp_seq=5 ttl=63 time=10.3 ms
64 bytes from 192.168.1.254: icmp_seq=6 ttl=63 time=4.45 ms
^C
--- 192.168.1.254 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 2.152/4.160/10.313/2.864 ms
cmnatic@CMNatic-THM-LPTOP:~$
```

The time taken for ICMP packets travelling between devices is measured by ping, such as in the screenshot below. This measuring is done using ICMP's echo packet and then ICMP's echo reply from the target device.

Pings can be performed against devices on a network, such as your home network or resources like websites. This tool can be easily used and comes installed on Operating Systems (OSs) such as Linux and Windows

Here we are pinging a device that has the private address of *192.168.1.254*. Ping informs us that we have sent six ICMP packets, all of which were received with an average time of 4.16 milliseconds.

Now you are going to do the same thing to ping the address of "8.8.8.8" on the deployable website in this task. Pinging the correct address will reveal a flag to answer the following question below.

```
user@thm:~$ ping -c 4
```

Enter the ip Address" **8.8.8.8**"

```
user@thm:~$ ping -c 4 8.8.8.8
```

Sent the ping request

```
user@thm:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=7.65 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=8.17 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=8.58 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=9.75 ms
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 8.132/9.428/10.957/1.057 ms
Flag: THM{I_PINGED_THE_SERVER}
```

We got the Flag

Quiz:

What protocol does ping use?

ICMP

✓ Correct Answer

What is the syntax to ping 10.10.10.10?

ping 10.10.10.10

✓ Correct Answer

What flag do you get when you ping 8.8.8.8?

THM{I\_PINGED\_THE\_SERVER}

✓ Correct Answer